



RANDOM VARIABLES AND DISTRIBUTIONS

Dr. Keerthy A S

Random Variables

- A random variable is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.
 - For example, if you roll a die, there are 6 possible outcomes
 - $S = \{1, 2, 3, 4, 5, 6\}$
 - $X = \text{Even number} \rightarrow \{2, 4, 6\}$
- Example: Consider random experiment of tossing a coin twice. Sample space is:

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

Define random variable X as the number of heads in the experiment:

$$X((T, T)) = 0, \quad X((H, T)) = 1,$$

$$X((T, H)) = 1, \quad X((H, H)) = 2$$



Probability distributions

- If we repeat an experiment many times, we can build up a distribution (table/formula) of the values it can take.
- Example : Tossing 3 coins together and noting down the number of heads that appear each time

Types of Random Variables

- ▮ **Discrete** random variables have a countable number of outcomes
 - ▮ Examples: getting a H on tossing a coin, Winning a competitive exam
- ▮ **Continuous** random variables have an infinite continuum of possible values.
 - ▮ Examples: blood pressure, weight, the speed of a car, the real numbers from 1 to 6.
- ▮ Example: Write down the probability distribution for sum of numbers turned up when 2 dice are rolled together

Discrete probability distributions

- A discrete probability distribution associates a probability with each value of a discrete random variable.
- Example:
 - Random variable has two values Rain/No Rain.
 - $P(\text{Rain}) = 0.2$, $P(\text{No Rain}) = 0.8$ gives a probability distribution.
- Properties:

$$0 \leq p(x_i) \leq 1$$

$$\sum_{x_i} p(x_i) = 1$$



The **cumulative distribution function** (CDF) of random variable X is defined as

$$F(x) = P(X \leq x), \text{ for all } x \in \mathbb{R}.$$

1. What's the probability that you roll a 3 or less in a single dice?
2. What's the probability that you roll a 5 or higher?

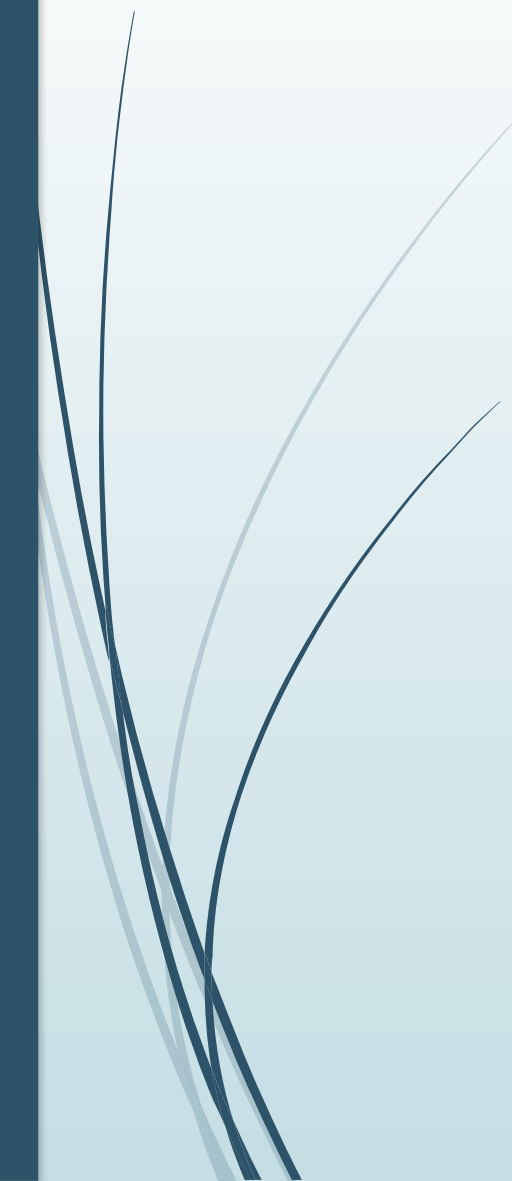
Probability Mass Function

- ▮ Probability mass function (pmf) is a statistical expression that defines a probability distribution (the likelihood of an outcome) for a discrete random variable
 - a. $f(x) = .25$ for $x = 9, 10, 11, 12$



b. $f(x) = (3-x)/2$ for $x=1,2,3,4$

c. $f(x) = (x^2+x+1)/25$ for $x=0,1,2,3$



An urn contains 6 red and 4 white balls. Three balls are drawn at random. Obtain the probability distribution of the number of white balls drawn.

X	0	1	2	3
P(x)	1/6	1/2	3/10	1/30

$$P(X = x) = \frac{\binom{4}{x} \binom{6}{3-x}}{\binom{10}{3}}$$

A random variable X has the following probability function:

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

- (i) Find k
- (ii) Evaluate $p(x \geq 6)$
- (iii) Evaluate $P(x \leq 6)$
- (iv) Evaluate $P(0 < x < 5)$

Continuous Random Variable

- ▮ A *continuous* random variable is one which can take *any* value in some range.
 - ▮ Example : crop yield, maximum temperature
- ▮ Since continuous random variables can take all values in a range, it is not possible to assign probabilities to individual values.
- ▮ We have a continuous curve, called a probability density function, which allows us to calculate the probability as a value within any interval.
- ▮ This probability is calculated as the area under the curve between the values of interest.
- ▮ The total area under the curve must equal 1.

Probability Density Function (PDF)

- X : **continuous** random variable
- $f(x)$: probability density function of X

➤ **Note:**

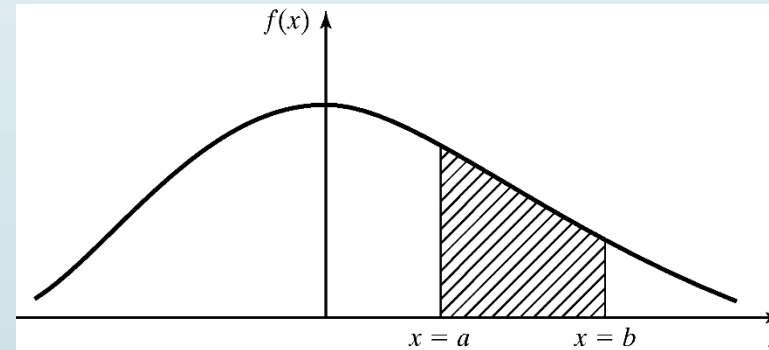
$$f(x) = \frac{d}{dx} F(x)$$

- $\mathbb{P}(X = x) = 0$
- $f(x) \neq \mathbb{P}(X = x)$
- $\mathbb{P}(x \leq X \leq x + \Delta x) \approx f(x)\Delta x$

➤ **Properties:**

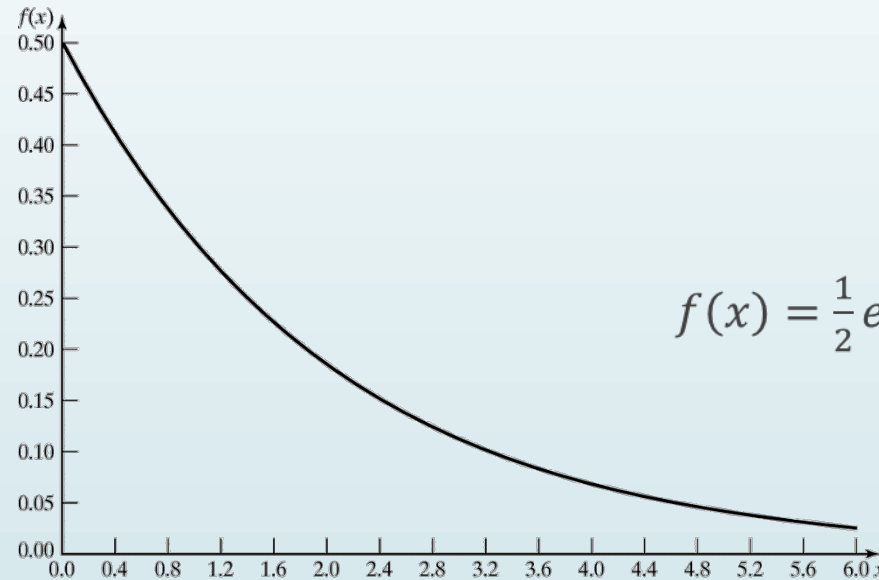
- $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx$
- $\int_{-\infty}^{+\infty} f(x) dx = 1$

CDF of X



Probability Density Function

Example: Life of an inspection device is given by X , a continuous random variable with PDF:



$$f(x) = \frac{1}{2}e^{-\frac{x}{2}}, \quad \text{for } x \geq 0$$

► Probability that the device's life is between 2 and 3 years:

$$\mathbb{P}(2 \leq X \leq 3) = \frac{1}{2} \int_2^3 e^{-\frac{x}{2}} dx = 0.14$$

Cumulative Distribution Function (CDF)

- X : **continuous** random variable
- $F(x)$: cumulative probability distribution function of X , or simply, probability distribution function of X

$$F(x) = \mathbb{P}(X \leq x)$$

- if X is continuous, then $F(x) = \int_{-\infty}^x f(t)dt$
- Properties
 - $F(x)$ is a non-decreasing function, i.e., if $a < b$, then $F(a) \leq F(b)$
 - $\lim_{x \rightarrow +\infty} F(x) = 1$, and $\lim_{x \rightarrow -\infty} F(x) = 0$
- All probability questions about X can be answered in terms of the CDF, e.g.:

$$\mathbb{P}(a < X \leq b) = F(b) - F(a), \text{ for all } a \leq b$$

Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{4x + k}{21} & \text{if } 3 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

Find the value of k .

□ Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{x}{8}, & 2 < x < 3, \\ \frac{1}{48}, & 3 < x < 36, \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(1 \leq x \leq 24)$.

▮ Let X be a continuous random variable with probability density function

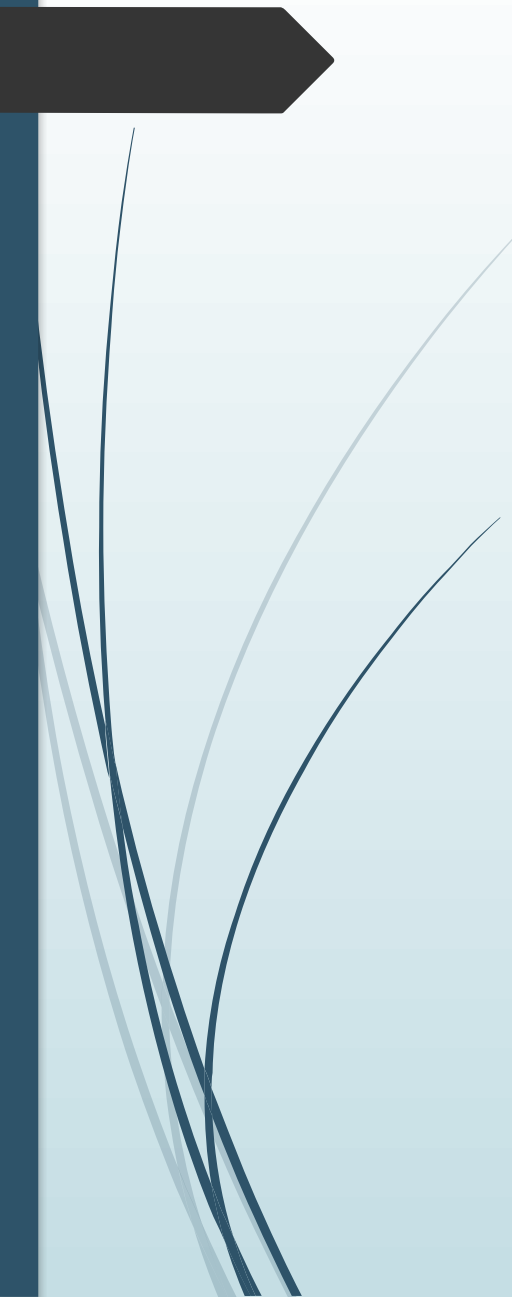
$$f(x) = \begin{cases} \frac{x+k}{54}, & 1 \leq x \leq 7, \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(X > 4)$

Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{2}{25}x + k, & 1 \leq x \leq 5, \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(3 \leq x \leq 4)$.



	Baseball	Basketball	Football	Total
Male	13	15	20	48
Female	23	16	13	52
Total	36	31	33	100

Joint probability density function of Discrete Random Variable:

if we have two random variables X and Y , and we would like to study them jointly, we define the **joint probability mass function** as follows:

$$p(x,y) = P[X = x, Y = y]$$

1. $0 \leq p(x, y) \leq 1$
2. $\sum_x \sum_y p(x, y) = 1$

Consider two random variables X and Y with joint PMF given

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{8}$
$X = 1$	$\frac{1}{8}$	$\frac{1}{6}$	$\frac{1}{6}$

- a. Find $P(X=0, Y \leq 1)$.
- b. Find the marginal PMFs of X and Y .
- c. Find $P(Y=1 | X=0)$.
- d. Are X and Y independent?



X	Y=1	Y=4
-2	0	1/4
-1	1/4	0
2	1/4	1/4

- a. Find $P(X=-1, Y \leq 4)$.
- b. Find the marginal PMFs of X and Y .
- c. Find $P(Y=1 | X=2)$.
- d. Are X and Y independent?

Marginal Density Function

Let X and Y denote two discrete random variables with joint probability function

$$p(x,y) = P[X = x, Y = y]$$

Then

$p_X(x) = P[X = x]$ is called the marginal probability function of X .

and

$p_Y(y) = P[Y = y]$ is called the marginal probability function of Y .

Conditional Density Function

Let X and Y denote two discrete random variables with joint probability function

$$p(x,y) = P[X = x, Y = y]$$

Then

$p_{X|Y}(x|y) = P[X = x|Y = y]$ is called the **conditional probability function** of X given $Y = y$

and

$p_{Y|X}(y|x) = P[Y = y|X = x]$ is called the **conditional probability function** of Y given $X = x$

Consider the following joint discrete probability mass function (pmf):

$p(x,y) = 0.1$ for $(x, y) = (1, 1), (2, 1), (1, 2),$ and $(2, 2)$

$p(x,y) = 0.2$ for $(x, y) = (0, 0)$

$p(x,y) = 0.4$ for $(x, y) = (3, 3)$

Obtain conditional pmf of X when $Y = 1$.

Consider two random variables X and Y with joint PMF given

	$Y = 2$	$Y = 4$	$Y = 5$
$X = 1$	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{24}$
$X = 2$	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{8}$
$X = 3$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$

- Find $P(X \leq 2, Y \leq 4)$.
- Find the marginal PMFs of X and Y .
- Find $P(Y = 2|X = 1)$.
- Are X and Y independent?

Joint probability density function of Continuous Random Variable:

Two random variable are said to have joint probability density function $f(x,y)$ if

$$1. \quad 0 \leq f(x, y)$$

$$2. \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$3. \quad P[(X, Y) \in A] = \int \int_A f(x, y) dx dy$$

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} x + cy^2 & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- a. Find the constant c .
- b. Find $P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2})$.

Marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \quad \text{for all } x,$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx, \quad \text{for all } y.$$

Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x, y) = \begin{cases} cx^2y & 0 \leq y \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c .
- Find marginal PDFs, $f_X(x)$ and $f_Y(y)$.
- Find $P(Y \leq \frac{X}{2})$.
- Find $P(Y \leq \frac{X}{4} | Y \leq \frac{X}{2})$.

For two jointly continuous random variables X and Y , we can define the following conditional concepts:

1. The conditional PDF of X given $Y = y$:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

2. The conditional probability that $X \in A$ given $Y = y$:

$$P(X \in A|Y = y) = \int_A f_{X|Y}(x|y)dx$$

Independent Random Variables:

When two jointly continuous random variables are independent, we must have

$$f_{X|Y}(x|y) = f_X(x).$$

Two continuous random variables X and Y are independent if

$$f_{XY}(x, y) = f_X(x)f_Y(y), \quad \text{for all } x, y.$$

Determine whether X and Y are independent:

$$f_{XY}(x, y) = \begin{cases} 8xy & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Expected Value and Variance

The **expected value** (or mean) of X , where X is a discrete **random variable**, is a weighted average of the possible **values** that X can take, each **value** being weighted according to the probability of that event occurring.

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- The expected value is obtained as follows:

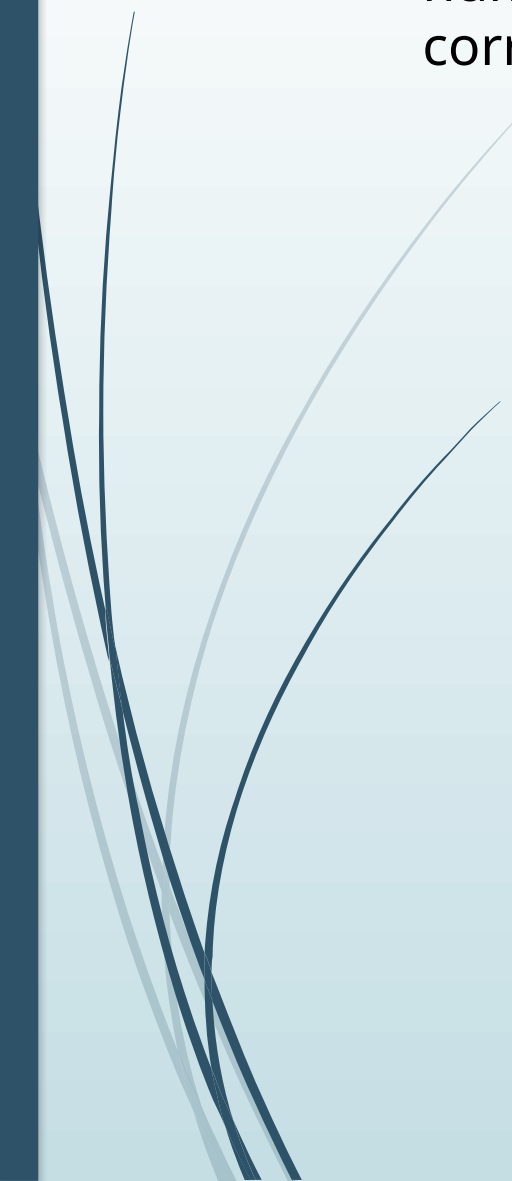
$$E(X) = \sum_{i=1}^n X_i p(X_i)$$

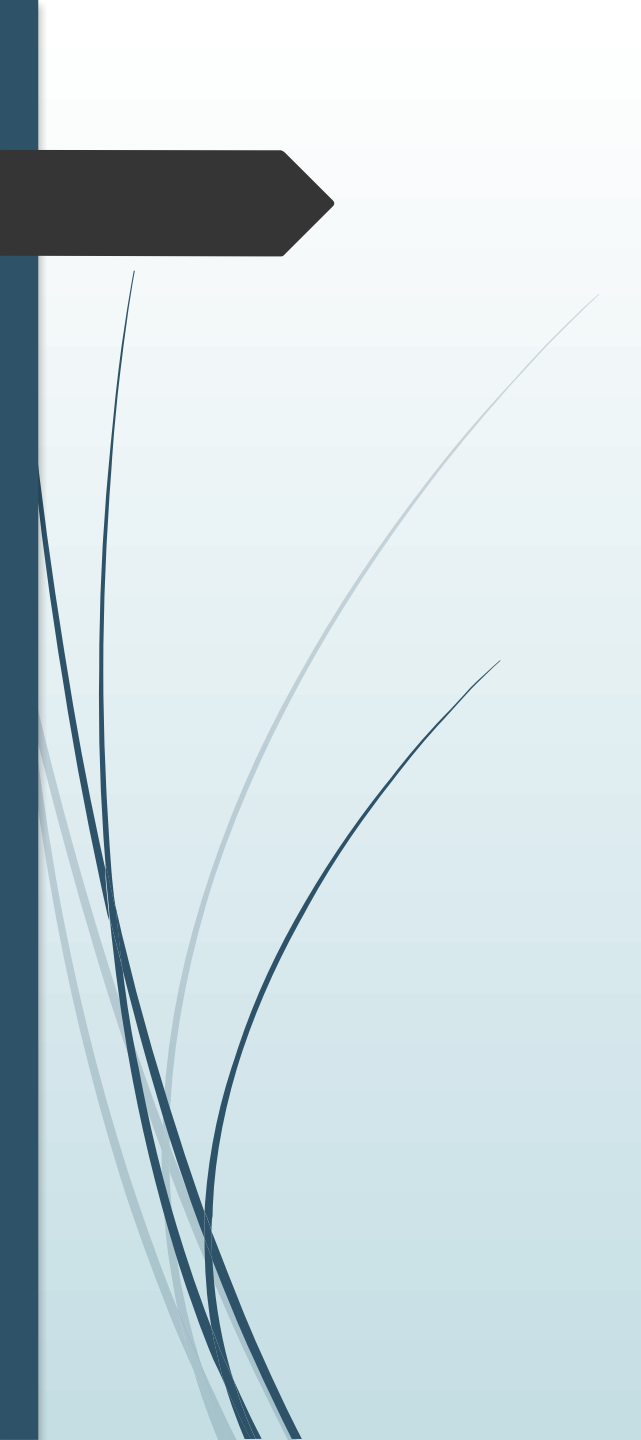
- $E(X)$ is the expected value of the random variable X
- X_i is the i -th possible value of the random variable X
- $p(X_i)$ is the probability that the random variable X will assume the value X_i

Expected value represents the long-run average outcome of the random experiment



Let an experiment consist of tossing a fair coin three times. Let X denote the number of heads which appear. Then the possible values of X are 0, 1, 2 and 3. The corresponding probabilities are $1/8, 3/8, 3/8$, and $1/8$. Find the expected value of X .





Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the expected value of X .



A random variable X has the following probability distribution. Compute expectation of X

X	-1	0	1	2
P(x)	1/3	1/6	1/6	1/3



Find the expected value of x .

$$f(x) = \frac{1}{x^3}, \quad 2 < x < 10$$

LAWS OF EXPECTED VALUE

- The laws of expected value are listed below:
 1. $E(c) = c$
 2. $E(cX) = cE(X)$
 3. $E(c' + cX) = c' + cE(X)$
 4. $E(X + Y) = E(X) + E(Y)$
 $E(X - Y) = E(X) - E(Y)$
 5. $E(XY) = E(X)E(Y)$, if X and Y are independent
- X and Y are random variables
- c, c' are constants
- $E(X), E(Y)$, and $E(c)$ are expected values of X, Y and c respectively.

Expected Value of a Function of a Discrete/Continuous Random Variable

$$E[g(X)] = \sum_{x_k \in R_X} g(x_k) P_X(x_k)$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} x + \frac{1}{2} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E(X^n)$, where $n \in \mathbb{N}$.

Variance

$$\text{Var}(X) = E[(X - \mu_X)^2] = EX^2 - (EX)^2.$$

$$\begin{aligned}\text{Var}(X) &= E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx \\ &= EX^2 - (EX)^2 = \int_{-\infty}^{\infty} x^2 f_X(x) dx - \mu_X^2\end{aligned}$$

Also remember that for $a, b \in \mathbb{R}$, we always have

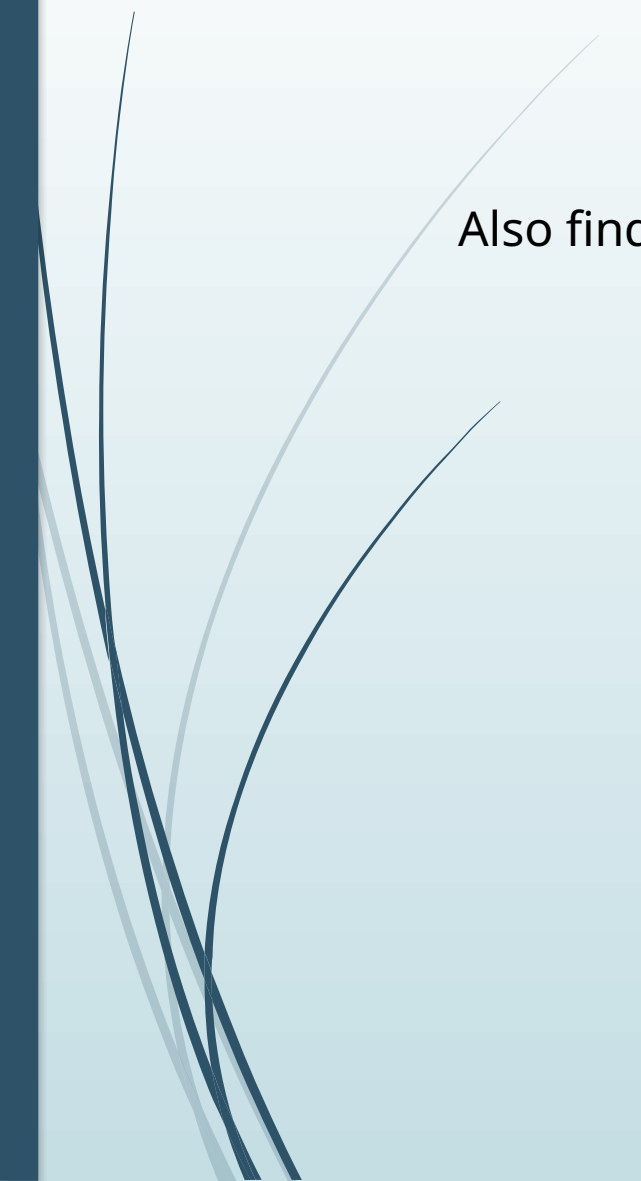
$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$



For the following probability distribution calculate the mean and variance of the random variable x

X	8	12	16	20	24
P(x)	1/8	1/6	3/8	1/4	1/12

Also find $E(2x+5)$



Moments and Moment Generating Functions

The ***n*th moment** of a distribution (or set of data) about a number is the expected value of the *n*th power of the deviations about that number.

- The *n*th moment of a distribution about zero is given by $E(X^n)$.
- The *n*th moment of a distribution about the mean is given by $E((X - \mu)^n)$.

- $Var(X) = E((X - \mu)^2) = E(X^2) - E(X)^2$.
- $Skew(X) = E((X - \mu)^3) = E(X^3) - 3E(X)E(X^2) + 2E(X)^3$
- $Kurt(X) = E((X - \mu)^4) = E(X^4) - 4E(X)E(X^3) + 6E(X)^2E(X^2) - 3E(X)^4$

Moment Generating Functions

A single expected value function whose derivatives can produce each of the required moments is called a **moment generating function**

$$M_X(t) = E(e^{tX}) = \sum_{\text{all } x} e^{tx} P(x)$$

$$M_X(t) = E(e^{tX}) = \int_x e^{tx} f(x) dx$$

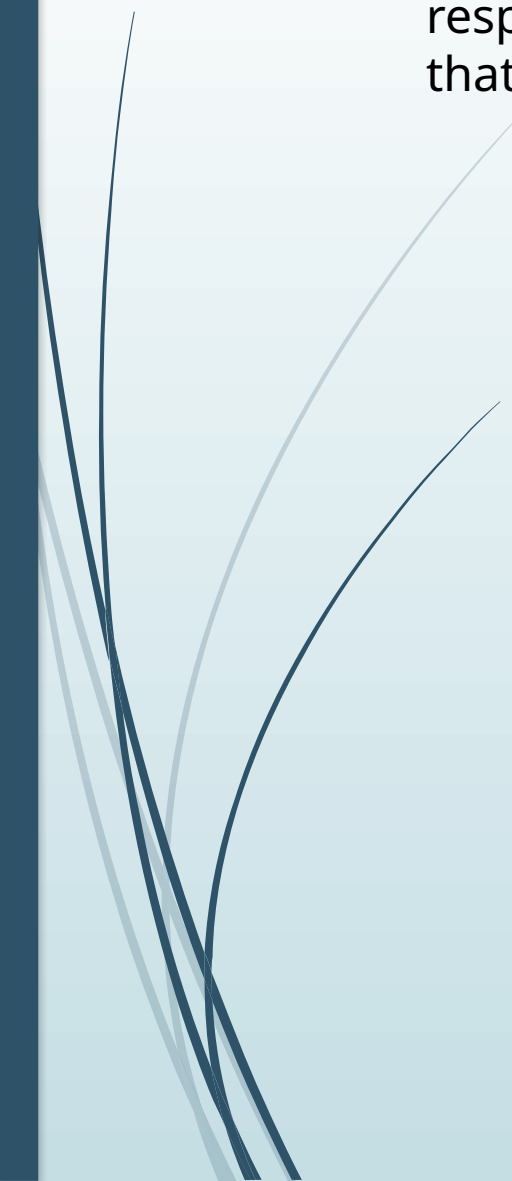
When the n th derivative (with respect to t) of the moment generating function is evaluated at $t=0$, the n th moment of the random variable x about zero will be obtained

$$E(X^n) = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}$$



Example – Discrete R V Distribution

Suppose you are given a 3 question multiple-choice test. Each question has 4 responses and only one is correct. Suppose you want to find the probability that you can just guess at the answers and get 2 questions right.



Binomial Distribution

A random variable X denoting the number of successes in an outcome of a Binomial experiment having n trials and p as the probability of success in each trial is called Binomial random variable. Its probability mass function is given by

$$P(X = x) = \begin{cases} nC_x p^x q^{n-x} & \text{for } x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases} \quad \text{where } q = 1 - p$$

where

n = the number of trials (or the number being sampled)

x = the number of successes desired

p = probability of getting a success in one trial

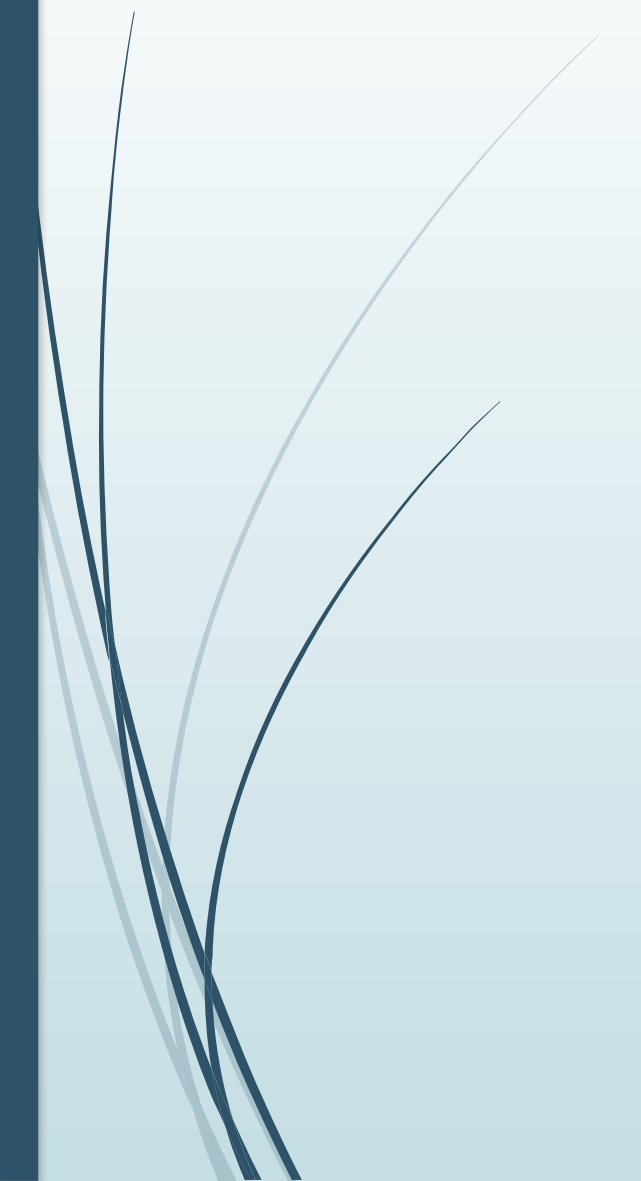
$q = 1 - p$ = the probability of getting a failure in one trial

Characteristics of $X \sim B(n, p)$

- ❖ **A discrete distribution.**
- ❖ **No of trials is pre-determined.**
- ❖ **Each trial must result in only two possible outcomes, ie either success (prob 'p') or failure (prob 'q' = 1- p).**
- ❖ **Prob of success in each trial must be same.**
- ❖ **Trials must be statistically independent.**



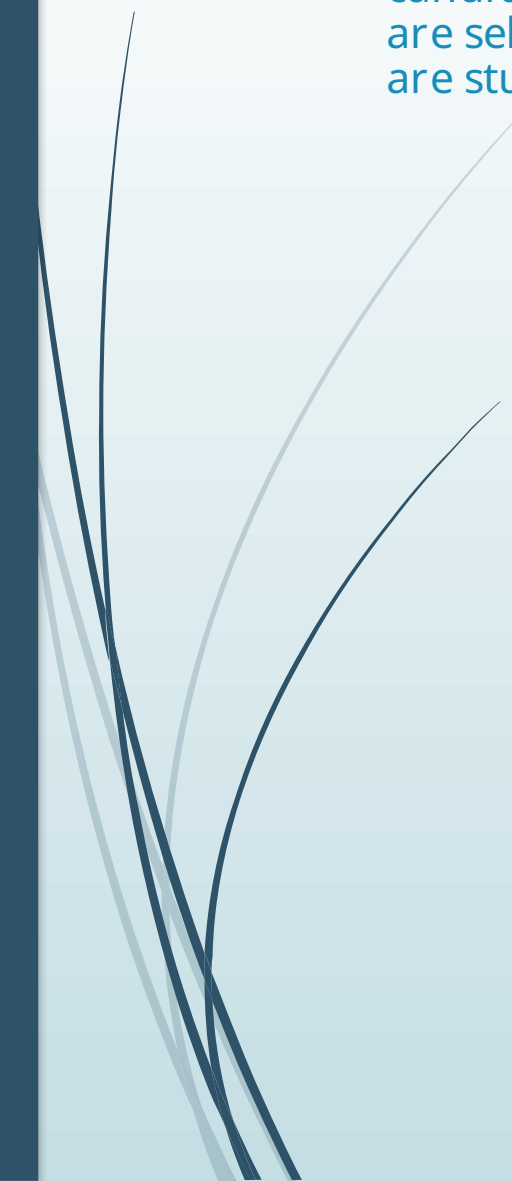
Mean & Variance of $X \sim B(n, p)$





QUESTION

Suppose that a central university has to form a committee of 5 members from a list of 20 candidates out of whom 12 are teachers and 8 students. If the members of the committee are selected at random, what is the probability that the majority of committee members are students?





An insurance salesman sells policies to five men, all of identical age and good health. According to the actuarial tables, the probability that a man of this particular age will be alive 30 years hence will be $\frac{2}{3}$. Find the probability that in 30 years

- (i) all five men
- (ii) at least one man
- (iii) at least 3 men
- (iv) at most 3 men, will be alive.

A dark grey arrow points to the right, and several thin, curved lines sweep upwards from the bottom left corner.

A student writes a five question multiple-choice quiz. Each question has four possible responses. The student guesses at random for each question. Calculate the probability for each possible score on the test from 0 to 5.

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Thirty percent of students at a large university are known to be short-sighted. If twenty students are picked at random, find the probability that at most two of them are shortsighted

The Poisson distribution

In statistics and probability, *the number of random, independent occurrences of the same event within a fixed interval* are frequently modeled using a *Poisson process*.

Examples:

- Suppose that, on average, 6 births occur at Hospital per day. Obviously, some days will have fewer births, and some days will have more. Given n , what is the probability that during a given day, Hospital assists in the birth of n babies?
- Approximately 0.56 meteorites weighing 10 grams or more strike the Earth every hour (on average). What is the probability no such meteorites fall to the Earth on a given hour?
- Suppose that trees in a forest are more or less evenly distributed, with an average of 720 trees per acre. What is the probability that a randomly chosen acre of the forest has ≥ 800 trees?

If one models this sort of situation carefully, one arrives at the *Poisson distribution*: A Poisson random variable X “with parameter λ ” ($\lambda > 0$), which counts events in a Poisson process, is a variable that can take nonnegative integer values $0, 1, 2, \dots$. It takes the value n with probability

$$\underbrace{\text{Prob}(X = n)}_{\text{probability } n \text{ events occur over one interval}} = \frac{\lambda^n e^{-\lambda}}{n!}$$

POISSON DISTRIBUTION

- This distribution gives the probability of the different possible number of occurrences of an event in a given time interval under certain conditions.
 - The events occur independently
 - The events occur at random
 - The probability of an event occurring in a given time interval does not vary with time

$$\sum_{n=0}^{\infty} \text{Prob}(X = n) = \sum_{n=0}^{\infty} \frac{\lambda^n e^{-\lambda}}{n!} = 1$$

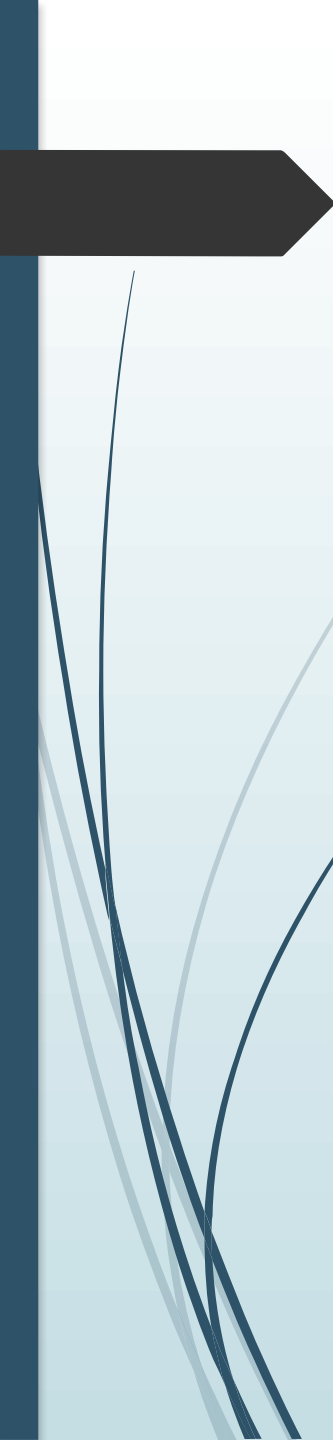
- The mean and variance of Poisson distribution = λ

Mean and Variance of Poisson Distribution

$$\begin{aligned}\text{Mean } E(X) &= \sum_{x=0}^{\infty} x p(x, \lambda) \\&= \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} \\&= \lambda e^{-\lambda} \left\{ \sum \left(\frac{\lambda^{x-1}}{(x-1)!} \right) \right\} \\&= \lambda e - \lambda(1 + \lambda + \lambda 2/2! + \dots) \\&= \lambda e - \lambda e \lambda \\&= \lambda\end{aligned}$$

$$\text{Variance } (X) = E(X^2) - E(X)^2$$

$$\begin{aligned}\text{Here } E(X^2) &= \sum_{x=0}^{\infty} x^2 p(x, \lambda) \\&= \sum_{x=0}^{\infty} x^2 p(x, \lambda) \\&= \sum_{x=0}^{\infty} \{x(x-1) + x\} p(x, \lambda) \\&= \sum_{x=0}^{\infty} \{x(x-1) + x\} \frac{e^{-\lambda} \lambda^x}{x!} \\&= e^{-\lambda} \sum x(x-1) \frac{\lambda^x}{x!} + \sum x e^{-\lambda} \frac{\lambda^x}{x!} \\&= \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda \\&= \lambda^2 e^{-\lambda} e^{\lambda} + \lambda = \lambda^2 + \lambda\end{aligned}$$



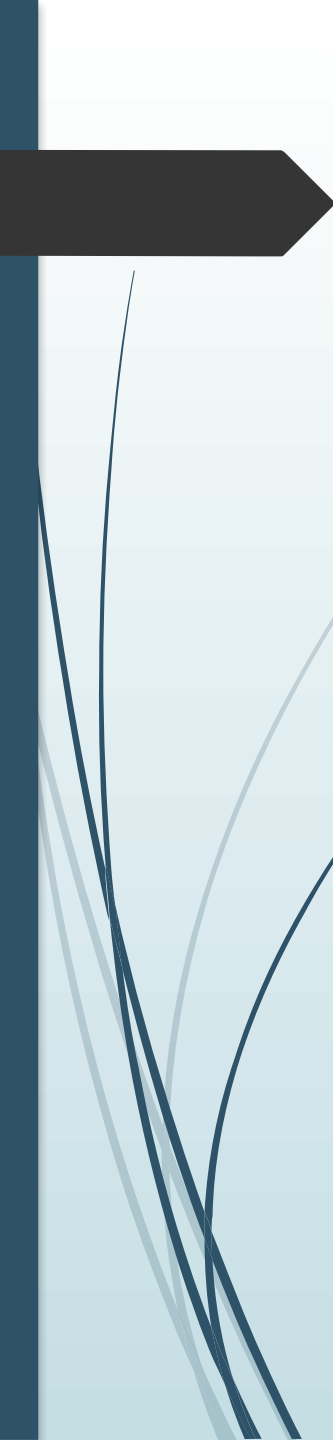
Between the hours 2PM and 4 PM , the average number of phone calls per minute coming into the switchboard of a company is 2.35. Find the probability that during one particular minute there will be at most 2 phone calls. Given $e^{-2.35} = 0.095374$



The number of industrial injuries per working week in a particular factory is known to follow a Poisson distribution with mean 0.5.

Find the probability that

- (a) in a particular week there will be:
 - (i) less than 2 accidents,
 - (ii) more than 2 accidents;
- (b) in a three week period there will be no accidents.



Sheep are randomly scattered in a field which is divided into equal size squares. There are three sheep on average on each square. Determine the probability that on a given square there will be ...

- a) no more than five sheep.
- b) more than two but no more than seven sheep.



The water from a lake is tested and is found to contain on average three bacteria per litre of water. A sample of 250 ml is collected from the lake.

a) Determine the probability that the 250 ml water sample will contain ...

i. ... exactly two bacteria.

ii. ... at least two bacteria.

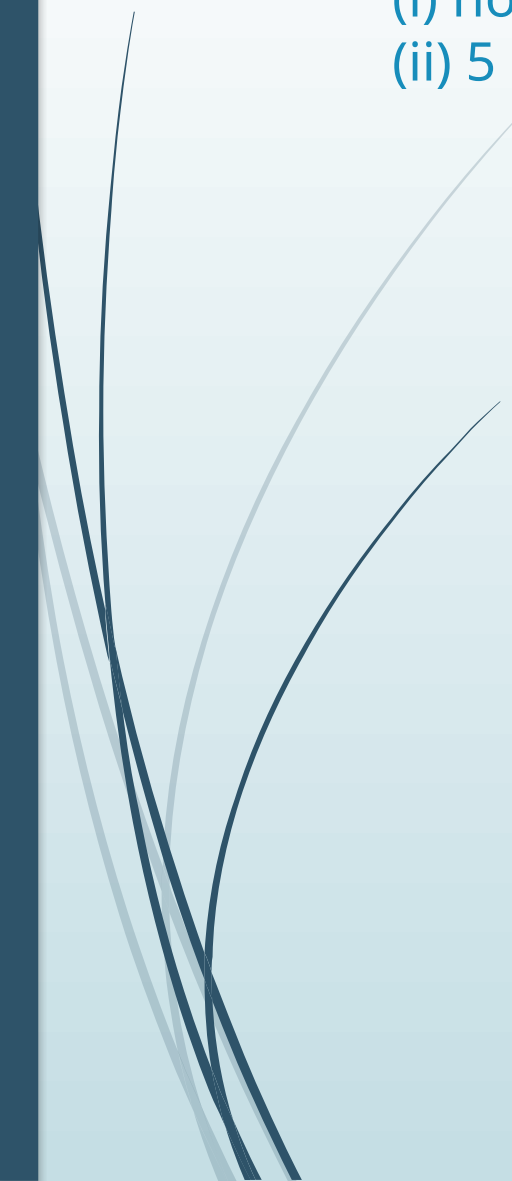
A larger sample of two litres of water is collected from the lake.

b) Find the probability that this larger sample will contain less than ten but no less than six bacteria.



If 5% of the electric bulbs manufactured by a company are defective, use Poisson distribution to find the probability that in a sample of 100 bulbs

- (i) none is defective,
- (ii) 5 bulbs are defective.



Example

In a certain part of the country 1 in 50 children have accidentally broken a bone in their body by the age of five. If 100 five year old children are chosen at random what is the probability that exactly 4 of the children have suffered a broken bone?

Using the binomial distribution

This distribution is a binomial distribution

with $n = 100$, $p = \frac{1}{50}$ and $q = \frac{49}{50}$.

The probability of 4 children having had broken bones is ${}^{100}C_4 \left(\frac{1}{50}\right)^4 \left(\frac{49}{50}\right)^{96} = 0.0902$.

In such cases the Poisson distribution can provide a good approximation to the binomial distribution. For this to be the case you require

- n to be large
- p to be small (and so the event is rare)
- np is not to be large (typically less than 10).

In this example, the mean of the binomial distribution is given by $np = 100 \times \frac{1}{50} = 2$

Poisson approximation, take $\lambda = 2$.

the probability of 4 children having suffered a broken bone = 0.0902.

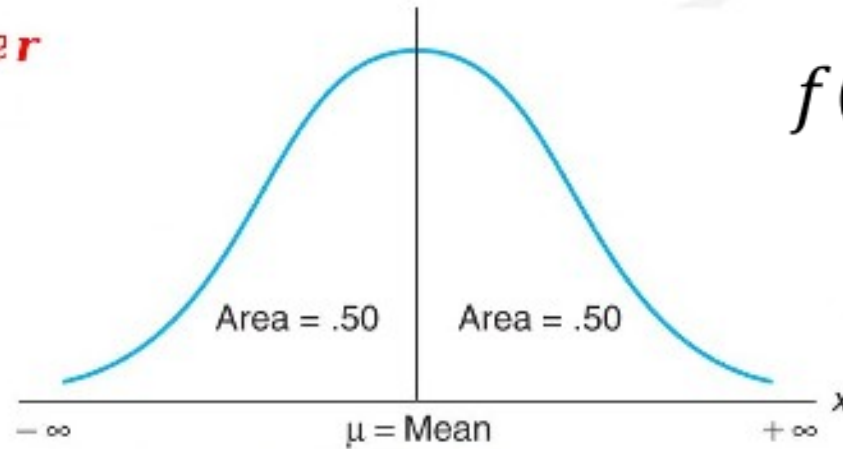


Fit a Poisson distribution and calculate the theoretical frequencies

x	0	1	2	3	4
f	123	59	14	3	1

Normal Distribution

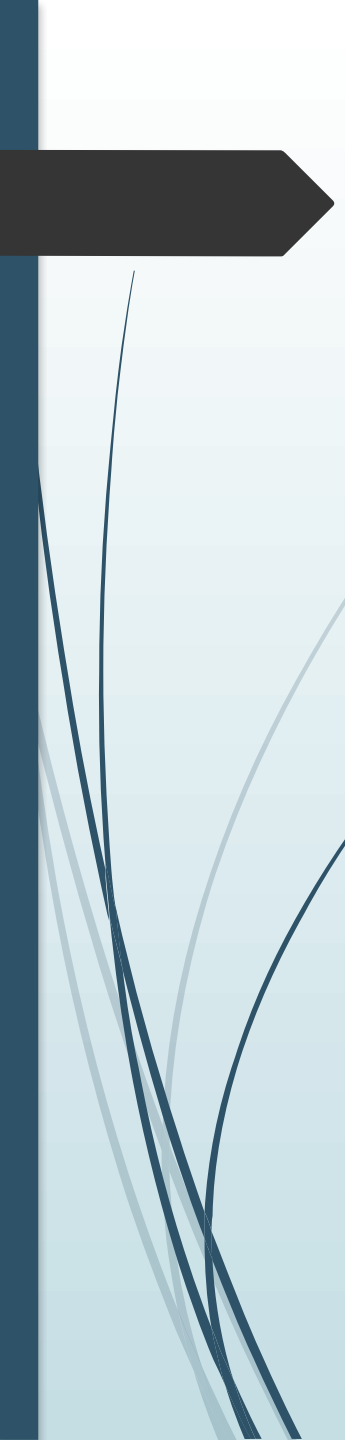
- The **normal distribution** is a continuous probability distribution that is symmetrical on both sides of the mean.
- The center of a normal distribution is its **mean** μ .
- The **area under probability, sums to one.**



The normal curve

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

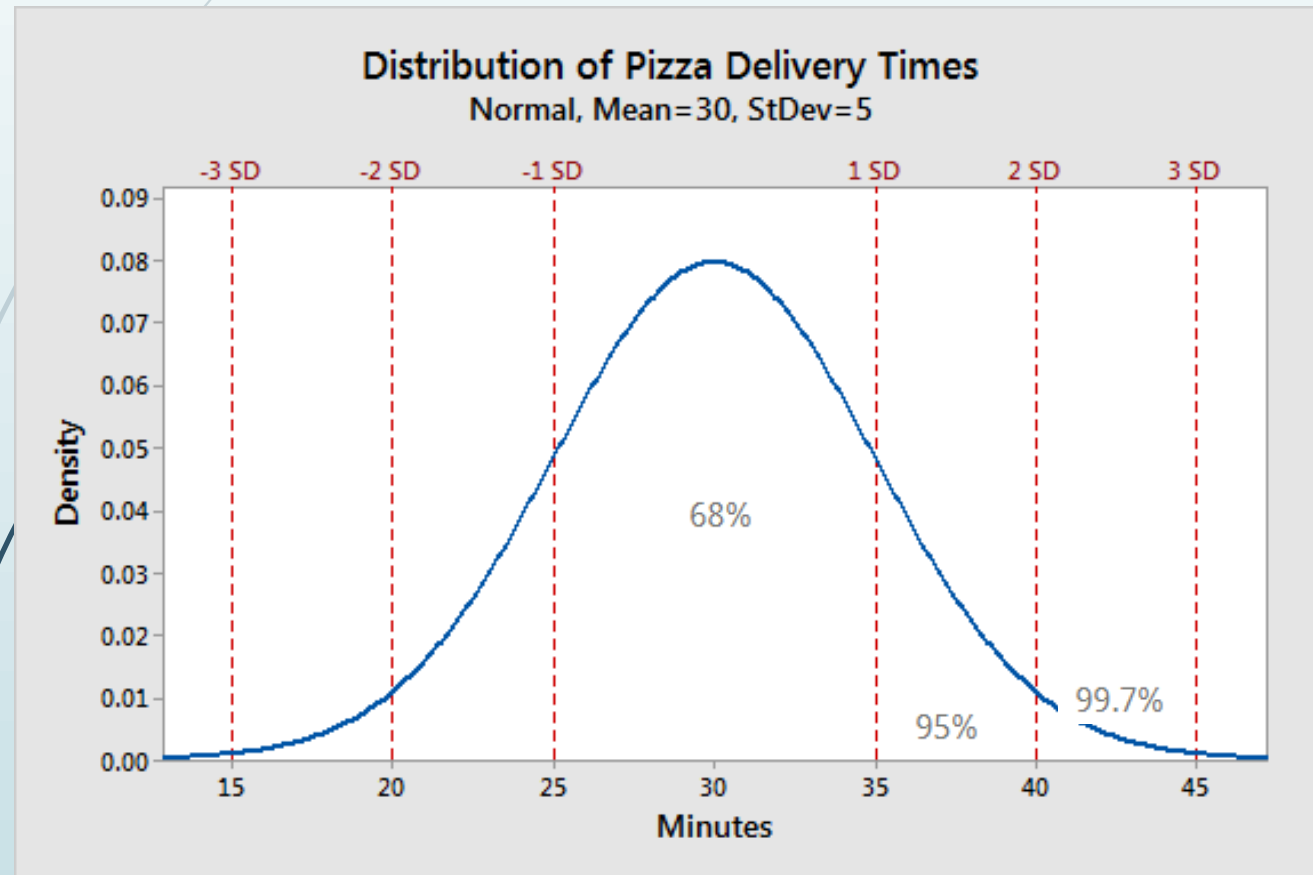
$$X \sim N(\mu, \sigma)$$



Characteristics of Normal Distribution:

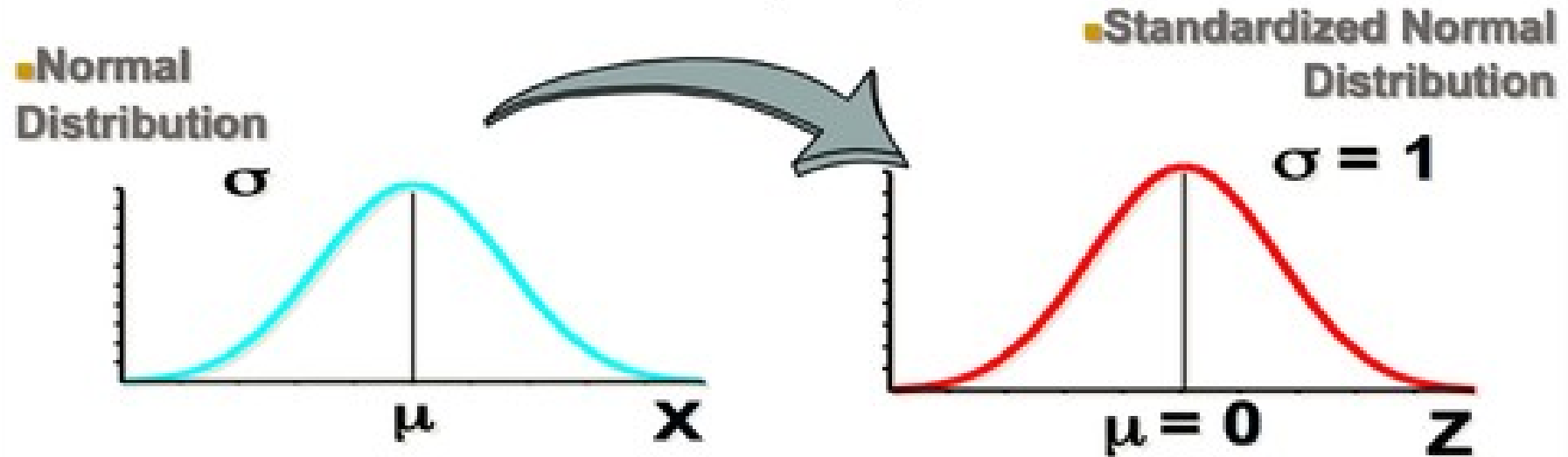
- 1) It is “**Bell-Shaped**” and has a single peak at the center of the distribution,
- 2) The arithmetic Mean, Median and Mode are equal.
- 3) The total area under the curve is 1.00; half the area under the normal curve is to the right of this center point and the other half to the left of it,
- 4) It is **Symmetrical** about the mean,
- 5) It is **Asymptotic**: The curve gets closer and closer to the X – axis but never actually touches it. To put it another way, the tails of the curve extend indefinitely in both directions.
- 6) The location of a normal distribution is determined by the Mean, μ , the Dispersion or spread of the distribution is determined by the Standard Deviation, σ .

Assume that a pizza restaurant has a mean delivery time of 30 minutes and a standard deviation of 5 minutes.



The Standard Normal Distribution

- A normal distribution with a mean of 0 and a standard deviation of 1 is called a standard normal distribution. It is the curve that results when any normal curve is converted to standardized scores and is written as: $Z \sim N(0, 1)$



- A normal curve with mean = 0 and standard deviation = 1 is called a standard normal curve.

- The process of *converting a value from a normal distribution to a value for the standard normal distribution* is called "**standardizing**" and requires the use of **z-scores**.
- z-scores tell us how far a value is from the mean

Z-score Calculation:

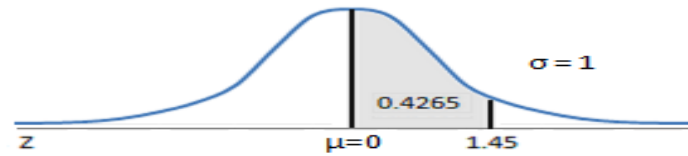
$$z = \frac{x - \mu}{\sigma}$$

- value being examined x
- population mean μ
- population standard deviation σ

Round z-scores to two decimal places.

Areas Under the One-Tailed Standard Normal Curve

This table provides the area between the mean and some Z score.
For example, when Z score = 1.45 the area = 0.4265.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

1) Find $P(Z \leq 1)$.

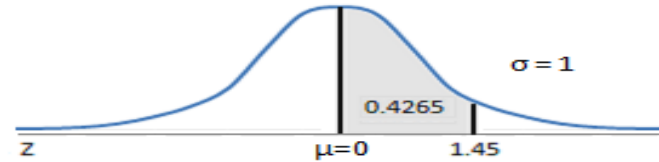
2) Find $P(Z \leq -2)$.

3) Find $P(Z \geq 2.5)$.

4) Find $P(Z \geq -1.5)$.

Areas Under the One-Tailed Standard Normal Curve

This table provides the area between the mean and some Z score.
For example, when Z score = 1.45 the area = 0.4265.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
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0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
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1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
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2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

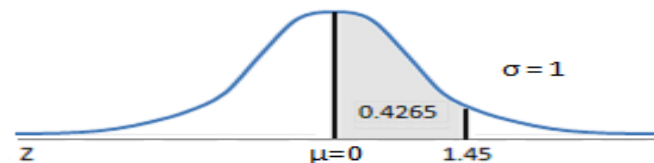
Find $P(-2 \leq Z \leq 2)$.

Find $P(-1.5 \leq Z \leq 2.5)$.

Areas Under the One-Tailed Standard Normal Curve

This table provides the area between the mean and some Z score.

For example, when Z score = 1.45 the area = 0.4265.



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

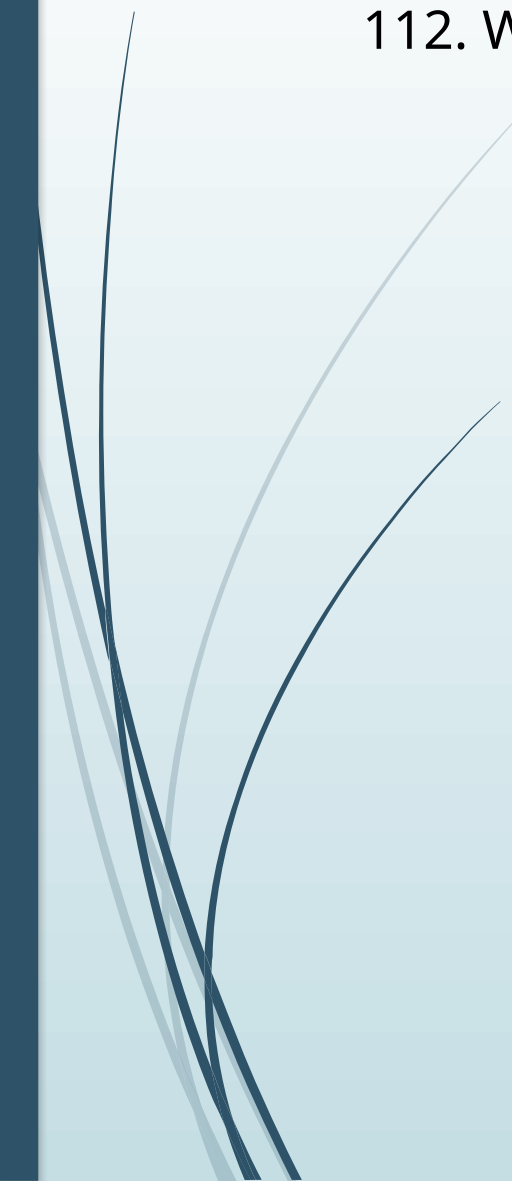
Find $P(-1.53 \leq Z \leq 2.16)$.

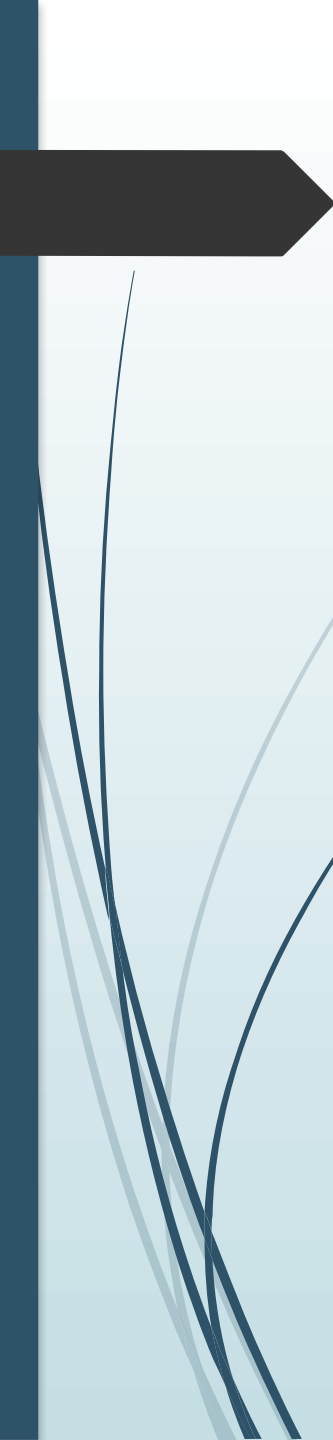
A dark grey arrow points to the right, and several thin, curved lines in shades of blue and grey sweep upwards from the bottom left corner.

If the length of newborn alligators, X , is normally distributed with mean $\mu = 6$ inches and standard deviation $\sigma = 1.5$ inches, what is the probability that an alligator egg about to hatch, will deliver a baby alligator between 4.5 inches and 7.5 inches?



Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT?





The average number of acres burned by forest and range fires in a large New Mexico county is 4,300 acres per year, with a standard deviation of 750 acres. The distribution of the number of acres burned is normal.

a) What is the probability that between 2,500 and 4,200 acres will be burned in any given year?

b) What number of burnt acres corresponds to the 38th percentile?

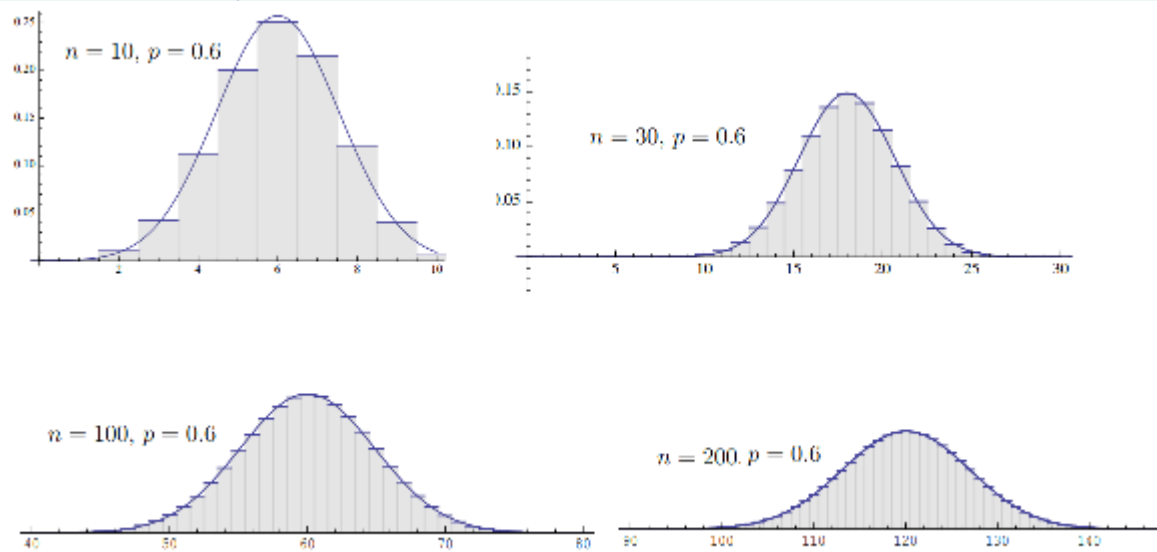
Binomial Approximation to Normal

Continuity correction

Given a binomial distribution X with n trials, success probability p , we can approximate it using a Normal random variable N with mean np , variance $np(1 - p)$.

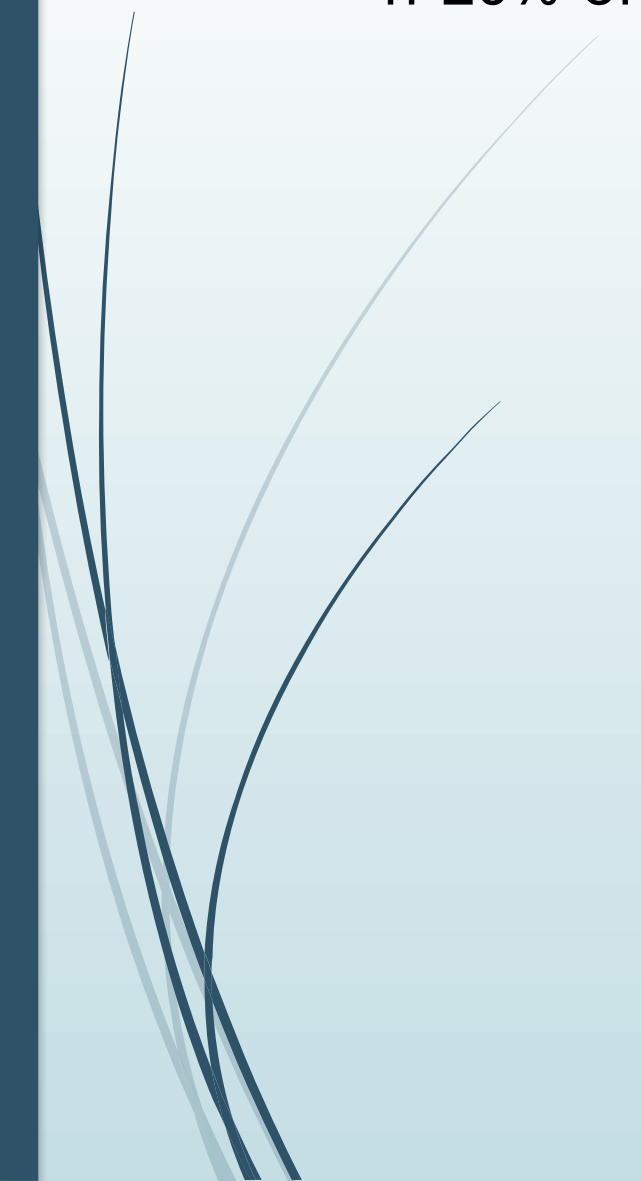
The continuity correction tells us that when we move from X to N , we should make the following changes to the probabilities we are calculating:

- ▶ $X \geq a$ changes to $N \geq a - 0.5$
- ▶ $X > a$ changes to $N \geq a + 0.5$
- ▶ $X \leq a$ changes to $N \leq a + 0.5$
- ▶ $X < a$ changes to $N \leq a - 0.5$



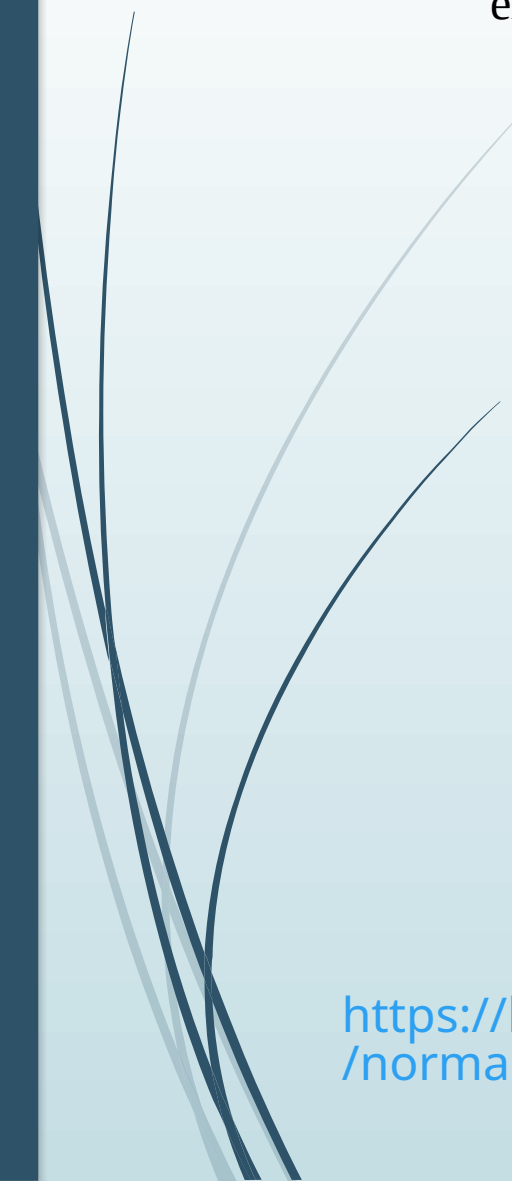


A normal distribution has 77 as mean. Find its standard deviation if 20% of the area under the curve lies to the right of 90.





A line for tickets to a local concert had an average (mean) waiting time of 20 minutes with a standard deviation of 4 minutes. If 2000 ticket buyers were in line, how many of them would expect to wait for more than 30 minutes?



<https://homepage.divms.uiowa.edu/~mbognar/applets/normal.html>

The Floppy Disk Company makes 3.5 inch floppy disks for disk drives that are 3.7 inches wide. The width of a manufactures disk is normally distributed with a standard deviation of 0.1 inches. The company manufactures 1000 disks every hour.

- a) 30% of the disks' widths were less than how many inches?
- b) 45% of the disks' widths were wider than how many inches?
- c) In one hour, how many disks would you expect to be between 3.45 inches and 3.55 inches?
- d) The middle 40% of floppy disks are between which two widths?